MIL-G-46003A(WC)
31 December 1976
SUPERSEDING
MIL-G-46003(ORD)
21 August 1961

MILITARY SPECIFICATION

GREASE, RIFLE

This specification is approved for use by Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of grease designed for the lubrication of rifles when used under conditions of sustained rain.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal		
L-P-349	-	Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate
VV-L-800	-	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rock Island Arsenal, ATTN: SARRI-RLE-S, Rock Island, Illinois 61201 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9150

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-595	-	Colors
FED-STD-791	-	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290	-	Packaging, Packing and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Standards

D128	-	Analysis of Lubricating Grease
D217	-	Test for Cone Penetration of Lubricating Grease
D270	-	Sampling Petroleum and Petroleum Products
D445	-	Test for Kinematic Viscosity of Transparent and Opaque Liquids (And the Calculation of Dynamic Viscosity)
D566	-	Test for Dropping Point of Lubricating Grease
D1264	-	Test for Water Washout Characteristics of Lubricating Greases
D1743	-	Test for Rust Preventive Properties of Lubricating Greases

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Grease furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.5.1 and 6.3). Any change in the formulation of a qualified product will necessitate its requalification.

3.2 Material. The grease shall be a mixture of mineral or synthetic oil and a jelling agent, with or without other added material, suitably stabilized to ensure that the finished product will meet the requirements of this specification.

3.3 Work penetration. The worked penetration of the grease shall be not less than 220 nor more than 270 when determined as specified in 4.6.

3.4 Dropping point. The dropping point of the grease shall be not less than 87.8°C (190°F) when determined as specified in 4.6.

3.5 Mineral oil viscosity. The viscosity of the mineral oil contained in the grease shall not be less than 108 cs. nor more than 172.6 cs. at 37.8°C (100°F) when determined as specified in 4.6.

3.6 Water resistance. When tested at 37.8 ± 2.8°C (100 ± 5°F) as specified in 4.6, not more than four percent by weight of the grease shall be washed from the bearing.

3.7 Rust prevention. When tested as specified in 4.6, the outer cup raceway and each roller shall not exceed a corrosion rating of 2.

3.8 Corrosion resistance. When tested in accordance with 4.6, the copper strip shall show no evidence of green color, pitting, or etching, nor shall any brown or black stain remain on the strip after washing with normal hexane as specified.

3.9 Storage stability. After testing as specified in 4.6.1, the grease shall meet all the requirements of this specification.

3.10 Performance. When tested as specified in 4.6.2, five rifles, 7.62MM., automatic, M14 lubricated with grease which has been subjected to the six months storage stability test (see 4.6.1), shall be wet fired using no less than 500 rounds of ammunition for each rifle. Freezing of one rifle bolt, or the occurrence of more than six malfunctions in the five rifles due to operation of the bolt shall be cause for rejection of the grease.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

3.11 Workmanship. The grease shall be homogeneous, smooth in texture, free of abrasives, lumps, granular particles, entrapped air, and any odor or rancidity, perfume or free alcohol.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous mixture of grease offered for acceptance in a single isolated container, or manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in ingredient materials.

4.2.2 Packaged lot. An indefinite number of one pound cans or smaller unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of grease manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling.

4.3.1 Sampling for examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.2 Sampling for tests. Each lot of grease shall be sampled in accordance with ASTM D270.

4.4 Inspection.

4.4.1 Inspection of material. Inspection shall be in accordance with Method 9601 of Fed. Test Method Std. No. 791.

4.4.2 Examination of filled containers. Examine samples taken in accordance with 4.3.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing and marking requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plant of MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests.

a. Qualification tests
b. Quality conformance tests

4.5.1 Qualification tests. Qualification tests shall consist of all the tests specified in this specification.

4.5.2 Quality conformance tests. The contractor shall furnish ten pounds of grease with each lot supplied under contract or order. Quality conformance tests shall be made on this sample of grease. Tests of individual lots which shall serve as a basis for Government acceptance shall consist of all test requirements specified in Section 3 with the exception of the storage stability and the performance tests specified in 3.9 and 3.10 respectively.

4.6 Test methods. Perform tests in accordance with applicable methods listed in Table I and in paragraphs 4.6.1 and 4.6.2.

Table I - Test Methods

	Fed. Std. No. 791 Method No.	ASTM Method No.
Penetration, worked		D217
Dropping point		D566
Grease analysis*		D128
Viscosity, kinematic		D445
Water resistance		D1264
Rust prevention		D1743
Corrosiveness	5309	

*The oil shall be extracted from the grease by this method for the oil viscosity determination.

4.6.1 Storage stability. Three one-quart, round, double compression, tin-plate cans shall be filled with the grease, leveled to approximately one-half inch from the top, and tightly sealed. The filled containers shall be stored in an oven maintained at 37.8 ± 1.7°C (100 ± 3°F) for a period of six months ± five days. At the end of the storage period, the grease shall be tested for compliance with the requirements listed in 3.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

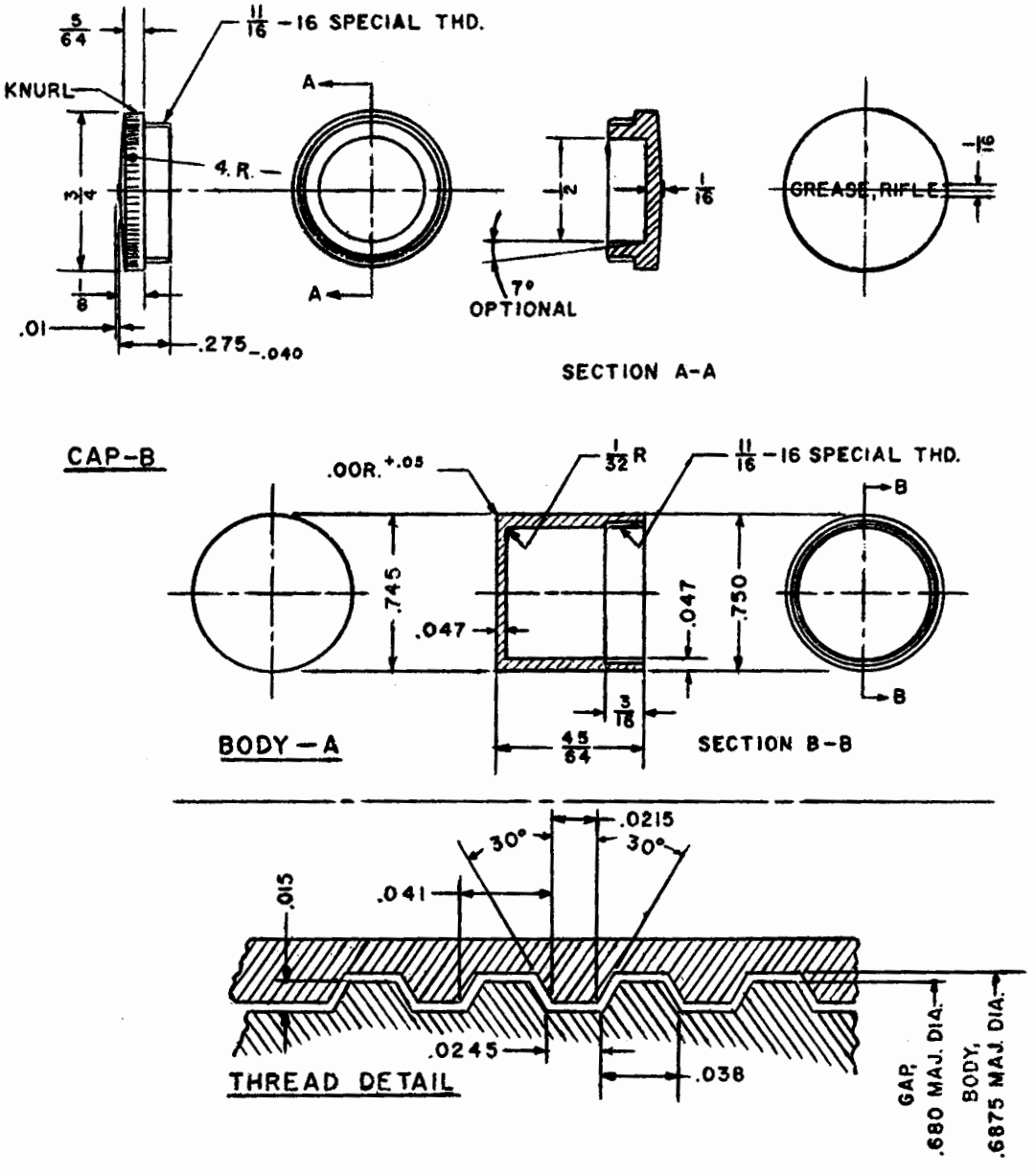


FIGURE 1: TUBULAR CONTAINER, PLASTIC, 2.5 cc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

4.6.2 Performance test.

4.6.2.1 Weapon preparation. Five rifles, 7.62MM, automatic, M14 (new or overhauled condition), shall be used for this test. Ambient temperatures of test shall be 24 ± 3°C (75 ± 5°F). M80 ball cartridges shall be used. The rifles shall be selected as having comparable firing rates from at least ten weapons. The rifles shall be thoroughly cleaned to remove traces of residues, lubricants and preservatives, then relubricated with oil conforming to VV-L-800. A light coating of rifle grease under test shall then be applied to the following surfaces: locking lugs of the bolt, operating lug and recesses, bolt guide, anti-friction roller on bolt, operating rod guide groove on the side of the receiver.

4.6.2.2 Test. Each weapon shall be mounted in turn on the functioning jack under the wet test apparatus (see figures 2 and 3) with the barrel depressed approximately 3°, or a sufficient amount to prevent accumulation of water in the rifle bore. The water mixing valve, which regulates the temperature of the water to be sprayed over the rifle, shall be adjusted to deliver water at 38 ± 1°C (100 ± 2°F.) at a rate of 1400 ± 200 ml. per minute.

The flow rate adjustment shall be made with a removable trough in place to prevent water from contacting the rifle until the temperature and flow rate are stabilized. The trough shall be removed and the water permitted to spray the receiver-bolt area of the rifle for five minutes with the bolt open (no magazine); then five minutes with the bolt closed, magazine in place and round chambered. With the water running, the rifle shall be fired 100 rounds in 20 round cycles as follows: 5 rounds semi-automatic; 3 to 5 rounds automatic in one burst; the remaining 10 to 12 rounds automatic in one burst. Magazines shall be changed without delay until the 100 rounds have been fired.

Any malfunction of the rifle shall be counted and recorded as to type and cause. The bolt shall then be closed, the hammer released and the bolt tested for effort required to open. If the bolt has not seized but can be unlocked by hand, the 100 round cycle shall be repeated starting with the 10 minute water spray (5 minutes bolt open-unloaded; 5 minutes loaded, bolt closed, chambered round). The firing shall be continued in this way until the bolt has seized or until 500 rounds have been fired. During the entire 500 round test, the rifle remains under the water spray. The five rifles shall be subjected, in turn, to the 500 round test.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified in the contract or order, preservation, packing and marking shall be in accordance with MIL-STD-290. The level of preservation shall be level A or C, and the level of packing shall be level A, B or C, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

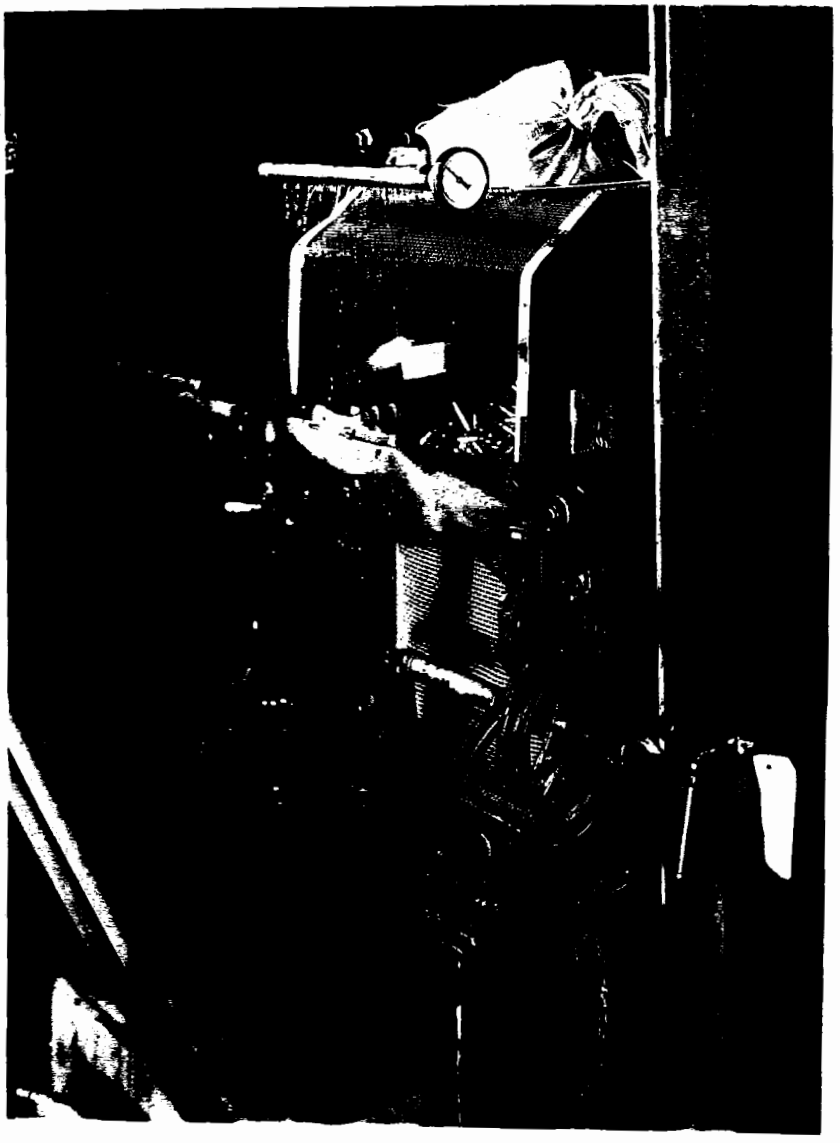


FIGURE 2: WET TEST FIXTURE FOR RIFLE PERFORMANCE TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

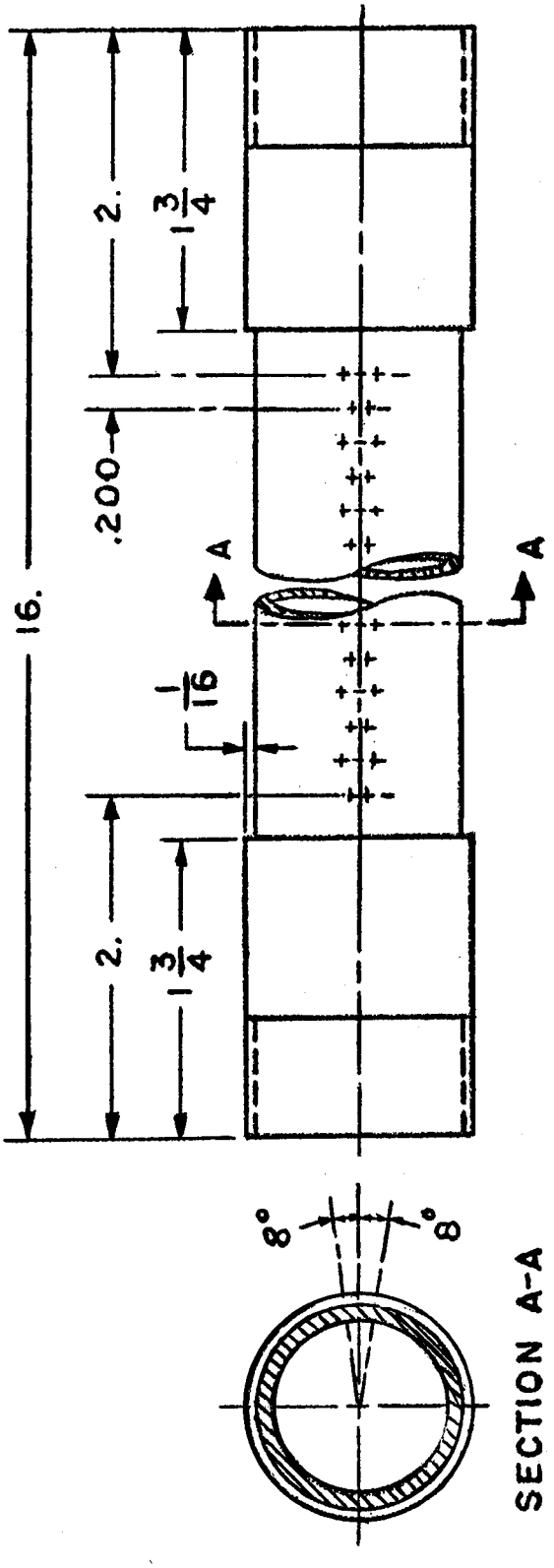


FIGURE 3: SPRAY HEAD FOR WET TEST FIXTURE
1" I.D. COPPER TUBING-HOLES 1/64 DIA.
PATTERNED 2-3-2-3---, 8° RADIALLY

NOTE: Dimensions are in inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

5.1.1 Container. Unless otherwise specified (see 6.2), the grease shall be contained in 2.5 cm^3 tubular containers which shall conform to the dimensions shown in Figure 1 for the body and cap. The body and cap shall be made of plastic conforming to type III, class MH of L-P-349. The body of the container shall be made of clear plastic. The cap shall be made of colored plastic, color number 13538 of Fed. Std. No. 595.

6. NOTES

6.1 Intended use. The grease covered by this specification is intended for use in the lubrication of rifles and other small arms only when such weapons are used under conditions of sustained rain. The temperature range under which this grease will be used is 2°C (35°F) to 38°C (100°F). This grease is not intended as a replacement for conventional oils under conditions other than sustained rain.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity of grease required.
c. Type and size of container (see 5.1.1 and 6.4).
d. Level of preservation and packing protection (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is U.S. Army Armament Command, General Thomas J. Rodman Laboratory, Rock Island Arsenal, ATTN: SARRI-RLR, Rock Island, Illinois, 61201, and information pertaining to qualification of products may be obtained from that activity.

6.4 National stock numbers. The following National Stock Numbers have been assigned to the grease covered by this specification:

2.5 cm^3 tube	9150-00-248-3480
1 lb. can	9150-00-754-0063

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.

Custodian:
Army - WC

Review activities:
Army - [redacted], ME, MR, MY
Navy - MC
DSA - GS, PS

User activity:
Army - PA

Preparing activity:
Army - WC

Project number:
9150-A402

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:38Z
Check the source to verify that this is the current version before use.